

TestLab 29119 — User Manual

Course: SENG 436 — Learner-as-Designer Project

Institution: Çankaya University

Standards: ISO/IEC 29119-4 (Test Techniques) · ISO/IEC 25010 (Quality Model)

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1. Introduction

TestLab 29119 is a browser-based educational serious game. You play the role of a **Test Manager** who must classify, apply, and select ISO/IEC 29119-4 test design techniques for realistic software projects. The game reinforces correct technique choice—not memorization of definitions alone—through four progressive phases and immediate feedback.

2. System Requirements

Requirement	Details
Browser	Recent Chrome, Microsoft Edge, or Firefox
JavaScript	Must be enabled
Internet	Required for Google Fonts (optional fallback to system fonts)
Storage	<code>localStorage</code> used for best score, leaderboard, and theme
Installation	None (web application)

3. How to Access the Game

Online

Live URL: <https://ismaiidogan.github.io/TestLab-29119/>

Local

```
npm install
```

```
npm start
```

Open `http://localhost:3000` in your browser.

4. Game Overview

Item	Description
Maximum score	500 points
Phases	4 + Results
Duration	Approximately 25–40 minutes (first playthrough)
Language	English

Phase summary

Phase	Name	Max points	Skill tested
1	Classify	110	Map techniques to Specification / Structure / Experience categories
2	Apply	195	Choose techniques for nine realistic scenarios
3	Strategize	110	Build a technique mix for three projects (Clause 4 constraints)
4	Test Cases	85	Select required EP and BVA test cases
—	Results	—	Grade, misconceptions, certificate, leaderboard

5. Getting Started

5.1 Welcome screen

1. Optionally enter **Your Name** and **School Name** (used on the leaderboard and certificate).
2. Click **Start Mission** to open the tutorial.
3. Defaults: Anonymous / Unknown School if fields are left empty.

5.2 Tutorial (six slides)

Slide	Topic
1	Welcome — Test Manager role
2	ISO/IEC 29119-4 — three technique categories
3	The three categories (Specification / Structure / Experience)
4	Clause 4 selection criteria
5	ISO/IEC 25010 link to Phase 2 scenarios

Slide	Topic
6	Your mission — overview of all four phases

- Use **Next** / **Back** and the dot indicator to navigate.
- Click **Skip** to go directly to Phase 1.
- On the last slide, **Start Phase 1** begins the game.

6. Phase-by-Phase Instructions

Phase 1 — Classify the Techniques

Goal: Drag each of **15 technique cards** into the correct category drop zone.

Zone	Category
Blue	Specification-Based
Green	Structure-Based
Red	Experience-Based

Actions:

- **Drag and drop** cards from the pool into a zone.
- **i** button — opens ISO detail modal (clause, description, best use).
- **Hint** button — reveals a hint; costs **4 points** per first use on each card.
- When all cards are placed, click **Check Answers**.
- Correct placements are highlighted in green; wrong cards show feedback with the correct category.
- Click **Continue to Phase 2** when finished.

Scoring: 6 points per correct card; +20 bonus if all 15 are correct; hint penalties subtracted from Phase 1 total (max 110).

Phase 2 — Apply to Scenarios

Goal: Answer **9 multiple-choice scenarios** by choosing the best test technique(s).

Features:

- Each scenario shows context, a question, and four options (A–D).
- **Keyboard:** Press **A**, **B**, **C**, or **D** to select; **Escape** closes modals.
- **ISO/IEC 25010 badge** on the scenario header links quality characteristics to the scenario.
- Some scenarios are **comparison mode** (e.g. EP vs BVA, Exploratory vs Ad Hoc).

- After answering, read **correct/wrong feedback** and any **misconception** note.
- Click **Next Scenario** until all nine are complete.

Scoring: 20–25 points per scenario (max 195 for Phase 2).

Phase 3 — Build Your Test Strategy

Goal: For each of **three projects**, select up to **five** test techniques that best fit the constraints.

Projects:

1. MedSafe — Healthcare (safety-critical)
2. QuickMVP — Startup food delivery (time/budget limited)
3. MigrateX — Legacy ERP migration (undocumented rules)

Actions:

- Read **constraints** (risk, budget, timeline, team, documentation).
- Click techniques to select/deselect (maximum 5).
- Hover for a quick preview; use the glossary (book icon) for ISO terms.
- Click **Submit Strategy** to see score, optimal selection, expert analysis, and key insight.
- Click **Next Project** or **Continue to Phase 4** on the last project.

Scoring: Up to 37 / 37 / 36 points per project (max 110). Partial credit applies; there is no single fixed algorithm—matching Clause 4 professional judgment.

Phase 4 — Write Your Test Cases

Goal: For **two exercises**, select all **required** EP and BVA test cases (checkboxes).

Exercises:

1. Login form — username field (5–20 characters)
2. Age field — insurance discount (18–65 inclusive)

Actions:

- Click rows to toggle selection.
- Click **Submit** to see coverage feedback (correct, missed, or unnecessary selections).
- Extra (non-required) selections incur a small penalty.
- Click **Next Exercise** or **See Final Results** on the last exercise.

Scoring: Up to 42 and 43 points (max 85). Each required item = 6 points when selected.

7. Scoring and Grades

Total score

Your **total** is the sum of all four phase scores (max **500**). The navbar **Score** updates as you play; **Best** stores your highest total in this browser.

Grade tiers (percentage of 500)

Grade	Minimum %	Title
ISO Master	90%	Outstanding mastery
Senior Tester	75%	Strong understanding
Test Engineer	60%	Good basics
Junior Tester	40%	Learning in progress
Trainee	0%	Review recommended

8. User Interface Reference

Navbar (top)

Control	Function
Top 10	Local leaderboard (this browser only)
Best	Highest total score saved locally
Score	Current run total
Book	ISO reference glossary (searchable)
Sun/Moon	Light / dark theme
Speaker	Mute / unmute sound effects

Phase progress bar

Shows four steps: **Classify** → **Apply** → **Strategize** → **Test Cases**. Active and completed steps are highlighted during gameplay.

9. Results Screen

After Phase 4 you will see:

- **Grade title** and emoji (e.g. ISO Master)
 - **Total points** and percentage
 - **Animated progress bar**
 - **Per-phase score breakdown**
 - **Misconceptions encountered** (learning points from wrong answers)
 - **Phase 3 key insights**
 - **Hint usage** summary (Phase 1)
 - **Download Certificate** — PNG file with your name, grade, and scores
 - **Play Again** — returns to welcome (scores reset for a new run)
-

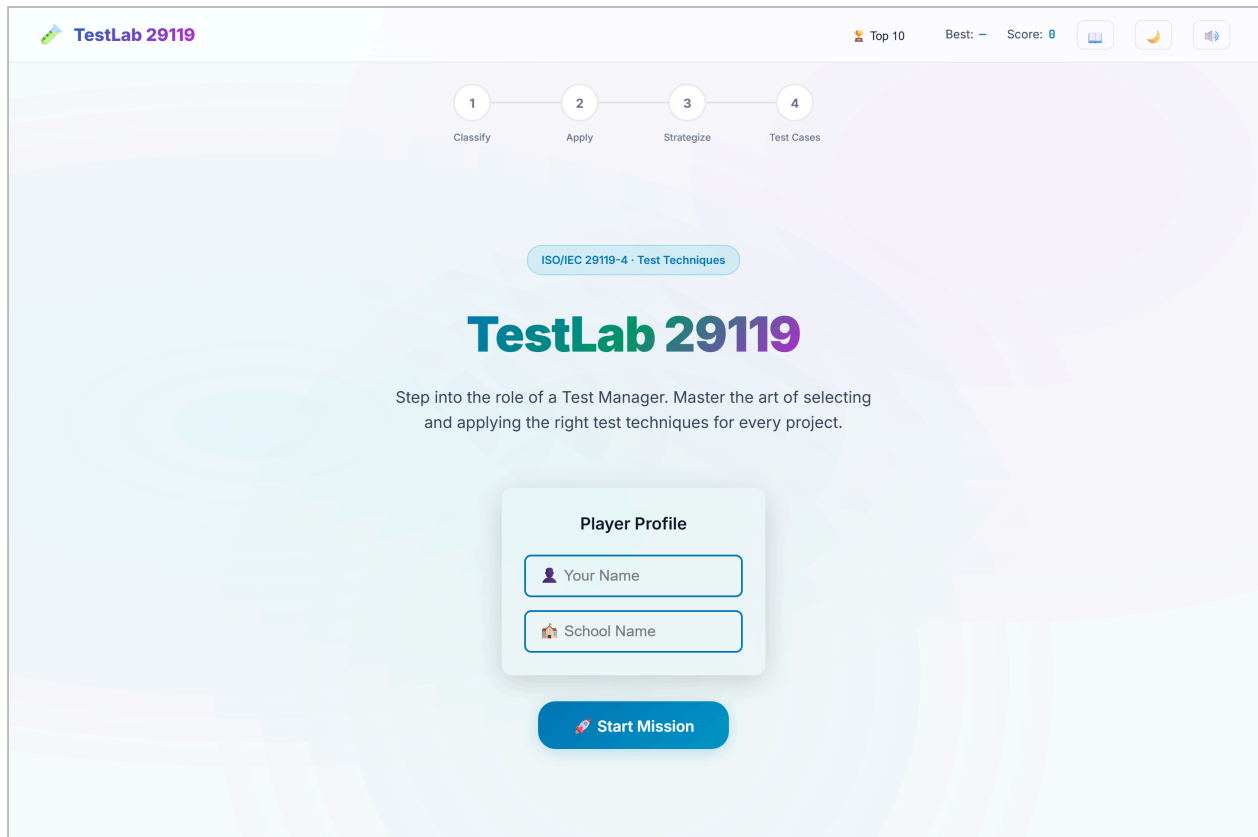
10. Troubleshooting

Problem	Solution
Score or leaderboard wrong	Data is per-browser; clear site data or use another browser
Fonts look plain	Check internet connection (Google Fonts CDN)
Drag-and-drop not working	Use a desktop browser; refresh the page
Game does not load	Ensure JavaScript is enabled; use the GitHub Pages URL or <code>npm start</code>

11. Appendix A — Screenshot Walkthrough

TestLab 29119 is a web application accessed via URL. The following screens document the user interface in play order.

1. Welcome — Player profile



The image shows a web interface for 'TestLab 29119'. At the top left is the logo 'TestLab 29119' with a test tube icon. At the top right are icons for a leaderboard (Top 10), a minus sign for 'Best', a score of 0, a book icon, a moon icon, and a speaker icon. Below these is a progress bar with four steps: 1. Classify, 2. Apply, 3. Strategize, and 4. Test Cases. A blue pill-shaped button contains the text 'ISO/IEC 29119-4 · Test Techniques'. The main heading is 'TestLab 29119' in large, bold, green and purple letters. Below it is a paragraph: 'Step into the role of a Test Manager. Master the art of selecting and applying the right test techniques for every project.' In the center is a 'Player Profile' form with two input fields: 'Your Name' (with a person icon) and 'School Name' (with a house icon). Below the form is a blue button with a rocket icon and the text 'Start Mission'.

TestLab 29119

Top 10 Best: — Score: 0

1 2 3 4
Classify Apply Strategize Test Cases

ISO/IEC 29119-4 · Test Techniques

TestLab 29119

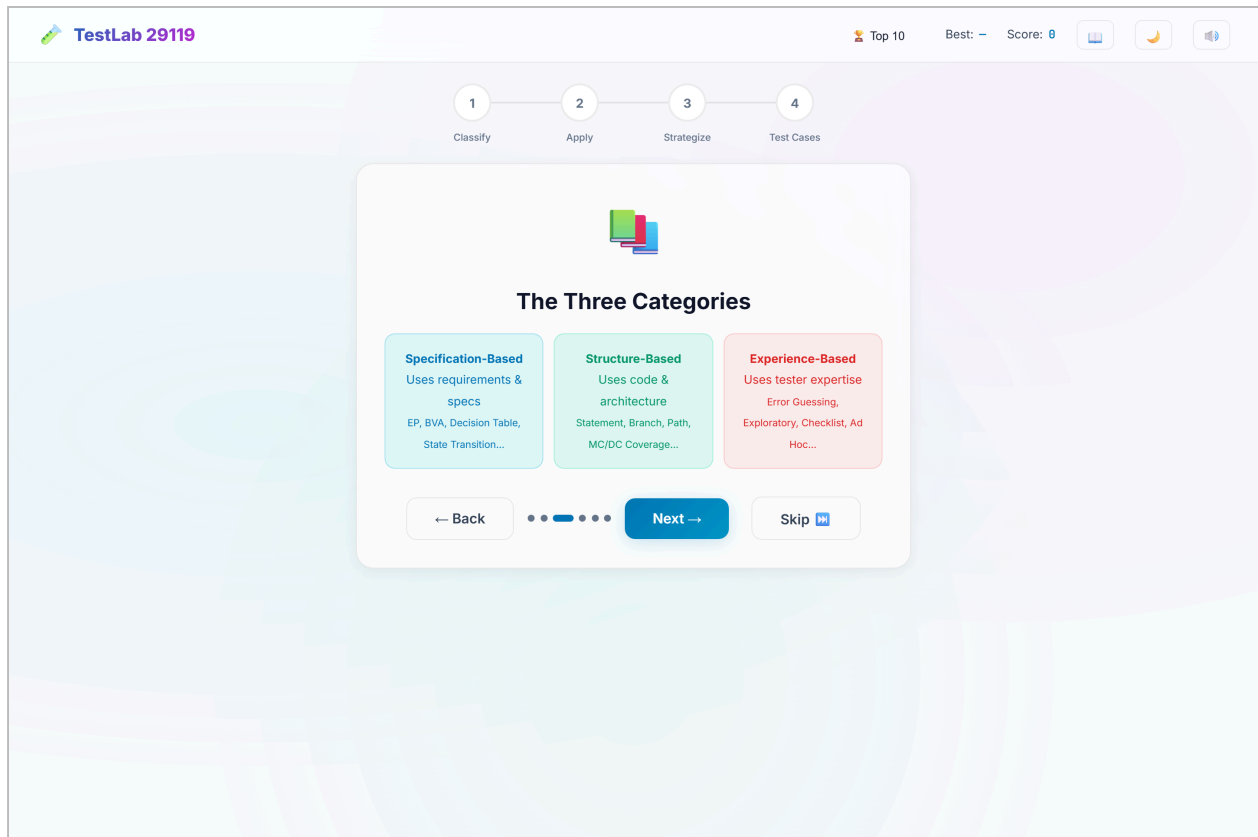
Step into the role of a Test Manager. Master the art of selecting and applying the right test techniques for every project.

Player Profile

Start Mission

Enter name and school, then **Start Mission**. Profile data appears on the certificate and leaderboard.

2. Tutorial — Three technique categories



Introduces Specification-Based, Structure-Based, and Experience-Based groups (ISO/IEC 29119-4).

3. Tutorial — Clause 4 selection criteria

The screenshot shows the TestLab 29119 interface. At the top, there's a header with the TestLab logo and ID, a progress bar with four steps (1. Classify, 2. Apply, 3. Strategize, 4. Test Cases), and user information (Top 10, Best: —, Score: 0). The main content area is a modal titled 'Selection Criteria (Clause 4)' with a gear icon. It explains that choosing the right technique depends on four criteria from Clause 4:

- Risk Assessment**: Product & project risk level
- Test Objectives**: Functional, safety, performance
- Test Levels**: Unit, integration, system, acceptance
- Organizational Context**: Time, budget, team expertise, docs

Below these, a warning states: 'There is NO fixed algorithm — technique selection is professional judgment.' At the bottom, there are navigation buttons: '← Back', a progress indicator with four dots (the second dot is highlighted), 'Next →', and 'Skip [icon]'.

Risk, objectives, test levels, and organizational context—the foundation for Phase 3.

4. Phase 1 — Classify

Classify

Apply

Strategize

Test Cases

PHASE 1

Classify the Techniques

Drag each test technique card to its correct category. 0 / 15 placed

Equivalence Partitioning

Boundary Value Analysis

Decision Table Testing

Checklist-Based Testing

Error Guessing

Path Coverage

Statement Coverage

Exploratory Testing

Classification Tree Method

Ad Hoc Testing

State Transition Testing

Use Case Testing

MC/DC Coverage

Branch Coverage

Pairwise Testing

■ Specification-Based

■ Structure-Based

■ Experience-Based

Drag fifteen technique cards into the three category drop zones.

5. Phase 2 — Scenario with ISO 25010 badge



Top 10Best: —Score: 0

1234

ClassifyApplyStrategizeTest Cases

PHASE 2

Apply to Scenarios

Choose the correct test technique for each real-world scenario. Scenario 1 / 9



Login Form Validation

ISO 25010 • Functional SuitabilityFunctional CorrectnessClauses 5.4 & 5.5

You are testing a login form that accepts a username (5-20 characters) and password (8-30 characters). The system should reject inputs outside these ranges and accept valid ones.

Which TWO techniques should you apply together for maximum effectiveness?

A Equivalence Partitioning + Boundary Value Analysis

B Statement Coverage + Branch Coverage

C Error Guessing + Ad Hoc Testing

D Path Coverage + MC/DC Coverage

Nine multiple-choice scenarios; each shows an ISO/IEC 25010 quality characteristic badge.

6. Phase 3 — Test strategy

9119

PHASE 3

Top 10Best: —Score: 0

Build Your Test Strategy

Given a project with real constraints, select the optimal technique mix. Project 1 / 3



MedSafe — Healthcare Patient Management System

A hospital is deploying a new patient management system that handles medication dosages, patient records, and appointment scheduling. The system is safety-critical — incorrect dosage calculations could harm patients.

**RISK ASSESSMENT**

CL. 4, 3

RISK

Very High (safety-critical, lives at stake)

**ORGANIZATIONAL CONTEXT**

CL. 4, 5

BUDGET

High (hospital budget allocated)

**ORGANIZATIONAL CONTEXT**

CL. 4, 5

TIMELINE

12 weeks for testing

**ORGANIZATIONAL CONTEXT**

CL. 4, 5

TEAM

5 experienced testers, 2 with medical domain knowledge

**ORGANIZATIONAL CONTEXT**

CL. 4, 5

DOCUMENTATION

Complete requirements specification available



Available Techniques

☐ Equivalence Partitioning

☐ Boundary Value Analysis

☐ Decision Table Testing

☐ State Transition Testing

☒ Statement Coverage

☒ Branch Coverage

☒ MC/DC Coverage

☐ Error Guessing

☐ Exploratory Testing

Selected: 0 / 5



Your Strategy

Select 5 techniques from the left panel

Submit Strategy

Select up to five techniques per project given risk, budget, timeline, team, and documentation constraints.

7. Phase 4 — Test case selection

TestLab 29119

Top 10Best: —Score: 0

1234

ClassifyApplyStrategizeTest Cases

PHASE 4

Write Your Test Cases

Select every test case you would include for the given scenario. Exercise 1 / 2

EP + BVA

Login Form — Username Field

A username field accepts 5–20 characters. The system must reject values outside this range. Select ALL test cases you would include using Equivalence Partitioning and Boundary Value Analysis.

EP Equivalence Partitioning | BVA Boundary Value Analysis | EXTRA Out of scope for EP/BVA

☐ EP EP — Valid partition: 10 chars (e.g. "tester1")

☐ EP EP — Invalid partition: 2 chars (below minimum)

☐ EP EP — Invalid partition: 25 chars (above maximum)

☐ BVA BVA — Boundary: 4 chars (just below min — expect FAIL)

☐ BVA BVA — Boundary: 5 chars (minimum valid — expect PASS)

☐ BVA BVA — Boundary: 20 chars (maximum valid — expect PASS)

☐ BVA BVA — Boundary: 21 chars (just above max — expect FAIL)

Choose required EP and BVA test cases via checkboxes.

8. Results — Score and grade

9119

Top 10Best: —Score: 0



Senior Tester

388 / 500 points (78%)

Phase 1 — Classify	95 / 110
Phase 2 — Apply	165 / 195
Phase 3 — Strategize	88 / 110
Phase 4 — Test Cases	40 / 85

Great work! You have a strong understanding of test technique selection.

No hints used — perfect independence!

Misconceptions Encountered

⚠️ EP and BVA are NOT mutually exclusive — they are complementary techniques designed to be used together.

Phase 3 Strategy Insights

- 1 Safety-critical systems demand the most rigorous techniques. MC/DC is often REQUIRED by safety standards, not optional.

Play Again

Download Certificate

Leaderboard

1. Alex Tester (Cankaya University)	388 pts
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Total score, grade tier, phase breakdown, misconceptions, and certificate download.

9. ISO reference glossary

TestLab 29119

Top 10Best:

ISO Reference

1

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388 / 500 points (78%)

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Search techniques & terms...

Ad Hoc Testing

Clause 7.5

Informal testing without predefined test cases, relying purely on tester intuition.

Boundary Value

Cl. 5.5

A value at the edge of an equivalence partition — where defects are statistically most likely to occur.

Boundary Value Analysis

Clause 5.5

Tests at the boundaries of equivalence partitions where defects are most likely.

Branch Coverage

Clause 6.3

Ensures every branch (decision outcome) in the code is executed at least once.

Checklist-Based Testing

Clause 7.4

Tests against a predefined list of items derived from experience and standards.

Classification Tree Method

Clause 5.6

Systematically identifies test-relevant aspects and their values using a tree structure.

Clause 4 — Selection Criteria

Cl. 4

Four factors for choosing techniques: Risk Assessment, Test Objectives, Test Levels, Organizational Context.

Decision Table Testing

Clause 5.7

Uses a table of input combinations and expected outputs to derive test cases.

Searchable glossary available from the navbar book icon during play.